

AI • MEDICAL IMAGING • DEEP LEARNING

Brain Tumor MRI Image Classification

Technical Documentation

Custom CNN and transfer learning, evaluated and deployed as a Streamlit application

4

tumour classes

2,443

MRI images

5

models compared

86.2%

best test accuracy

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VERSION	1.0
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REPOSITORY	https://github.com/13msrajput/Brain-Tumor-MRI-Classification
LIVE APP	https://brain-tumor-mri-classification-system.streamlit.app

For education and demonstration only. Not a medical device. Not for clinical use.

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1. Executive Summary

What was built, how well it works and what to keep in mind.

Brain tumours are not a single disease: the tumour type decides the treatment plan, and identifying it from an MRI scan is slow, needs specialist expertise and is not available everywhere. This project builds a deep learning system that reads a brain MRI image and classifies it into one of four categories: **Glioma**, **Meningioma**, **Pituitary tumour** or **No Tumor**.

A custom Convolutional Neural Network (CNN) built from scratch was compared with four ImageNet-pretrained networks (MobileNetV2, ResNet50V2, InceptionV3 and EfficientNetB0) using identical data, training controls and evaluation code. The best model was selected on **validation** macro F1-score, so the test set remained unseen until the final report. The selected model is served through a Streamlit web application that returns the predicted tumour type, the confidence, the probability of every class and a Grad-CAM heatmap showing which regions influenced the decision.

86.18% ResNet50V2 test accuracy	0.855 macro F1-score (test)	98.98% tumour sensitivity	34 / 246 test images misclassified
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Key results

Table 1. Headline results on the 246-image test set.

Model	Test accuracy	Test F1 (macro)	Tumour sensitivity	Role
ResNet50V2	86.18%	0.8546	98.98%	Selected (best validation F1)
InceptionV3	84.15%	0.8358	98.48%	Runner-up
MobileNetV2	82.52%	0.8232	99.49%	Best lightweight option
Custom CNN	78.46%	0.7530	91.88%	From-scratch baseline
EfficientNetB0	69.92%	0.6874	99.49%	Weakest on this dataset

Main findings

- **Transfer learning clearly helps.** The three strongest pretrained models (ResNet50V2, InceptionV3, MobileNetV2) all beat the custom CNN, which reached 78.46% test accuracy against 86.18% for ResNet50V2. EfficientNetB0 was the exception on this dataset (69.92%).
- **Meningioma is the hardest class.** Recall for meningioma is 68.3% with ResNet50V2; 16 of 63 meningioma scans were predicted as pituitary tumours, a confusion that appears in all three strong pretrained models.
- **Missing a tumour is rare.** ResNet50V2 labelled only 2 of 197 tumour scans as "No Tumor" (tumour sensitivity 98.98%).
- **The reported scores are probably optimistic.** About 19% of validation and 23% of test images share a source-scan ID with a training image (see Section 3.3), so results should be read as a strong demonstration, not clinical validation.

2. Project Objectives and Scope

Why the project exists and what it delivers.

2.1 Problem statement

Develop a deep learning solution that classifies brain MRI images into multiple categories according to tumour type, enhance it with transfer learning, and deploy a user-friendly web application for real-time predictions from uploaded scans.

2.2 Intended use cases

Table 2. Business use cases from the project brief.

Use case	How the system supports it
AI-assisted medical diagnosis	Gives radiologists a fast second reader that classifies scans and can reduce turnaround time.
Early detection and triage	Flags scans with a tumour pattern so specialists can review high-risk cases first.
Research and clinical trials	Groups patient datasets by tumour type to support studies and recruitment.
Second-opinion systems	Supports telemedicine and remote consultation where specialists are scarce.

2.3 Deliverables

Table 3. Project deliverables and where to find them.

Deliverable	Status	Location
Trained models (custom CNN and 4 pretrained, .h5)	Complete	models/ (generated by the notebook)
Training, evaluation and comparison notebook	Complete	Brain_Tumor_MRI_Classification.ipynb
Streamlit application	Complete	app.py
Model comparison	Complete	Section 6 of this document and model_comparison.csv
Public GitHub repository with README	Complete	github.com/13msrajput/Brain-Tumor-MRI-Classification
Clean, modular, commented code	Complete	Notebook (line-by-line comments) and app.py

3. Dataset

Source, structure, quality checks and a finding about overlap between splits.

3.1 Source and structure

The data is the *Labeled MRI Brain Tumor Dataset* (Roboflow Universe, licence CC BY 4.0). It contains **2,443** MRI images, all **640 x 640 pixels in RGB mode**, already divided into training, validation and test folders. Each folder holds one sub-folder per class, and a `_classes.csv` file lists the one-hot label of every image.

Table 4. Images per class and split.

Class	Train	Valid	Test	Total	Share
Glioma	564	161	80	805	32.95%
Pituitary	438	118	54	610	24.97%
Meningioma	358	124	63	545	22.31%
No Tumor	335	99	49	483	19.77%
Total	1,695	502	246	2,443	100%

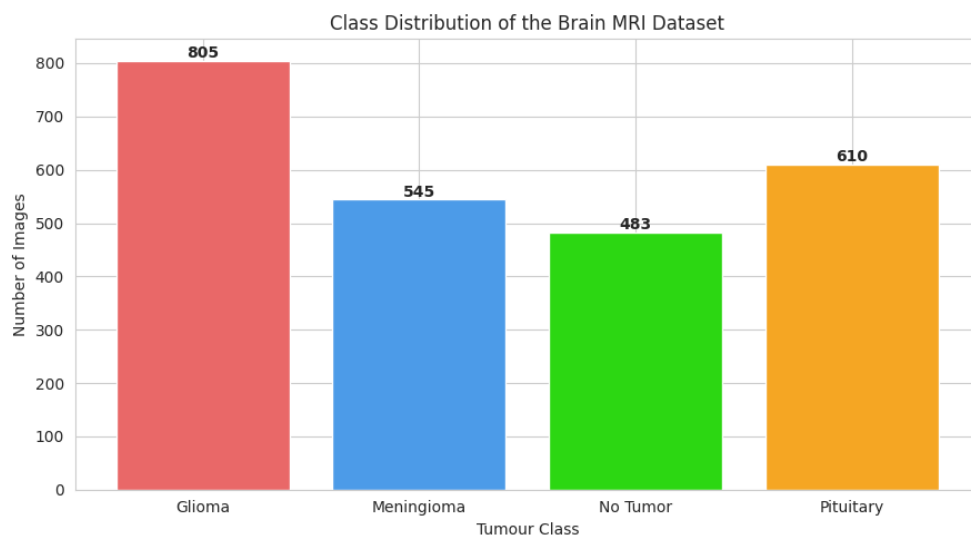


Figure 1. Class distribution of the dataset. Glioma is the largest class; the imbalance ratio is 1.67.

3.2 Data quality checks

Table 5. Validation of every image and label before modelling.

Check	Result
Missing values in the image table	0
Corrupted or unreadable images	0
Exact duplicate images (md5 fingerprint)	0
Folder name vs _classes.csv label mismatches	0 (all 2,443 images agree)
Distinct image resolutions	1 (640 x 640, RGB)
Class imbalance ratio (largest / smallest class)	1.67 (mild)

No cleaning step was needed, so the effort could focus on preprocessing and modelling. The mild class imbalance is handled with class weights and by reporting macro-averaged metrics next to accuracy.

3.3 Overlap between splits

File names contain the ID of the original scan (for example Tr-gl_0022). Comparing these IDs shows that **97 of 502 validation images (19.3%)** and **57 of 246 test images (23.2%)** share a scan ID with a training image. These are most likely augmented copies of the same scan created by the dataset provider before the split was made.

Why this matters

The official split was kept so that results stay comparable with the project brief. However, scores on the validation and test sets are probably somewhat higher than they would be on completely new patients. A patient-level split would give a more honest estimate.

4. Exploratory Data Analysis

Nine charts were produced in the notebook; the most informative ones are summarised here.

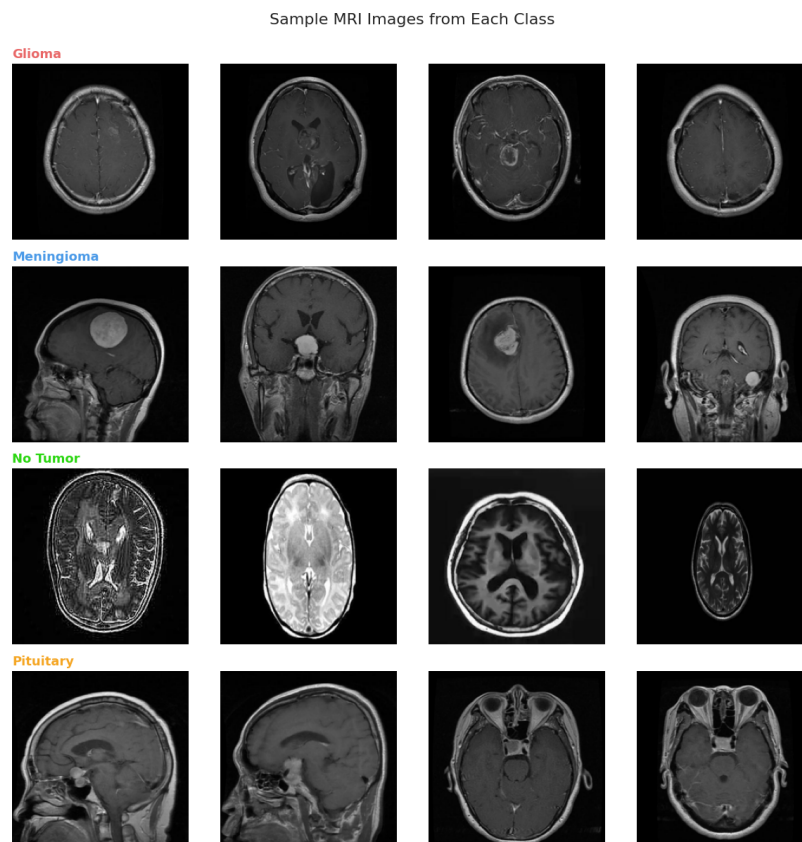


Figure 2. Random MRI samples of each class. Scans come from axial, sagittal and coronal planes and from different scan settings.

- **High visual variety.** Viewing plane and scan style differ strongly, so the model must learn local patterns rather than rely on one fixed layout.
- **Gliomas** appear mostly as axial slices with a darker, uneven mass; **meningiomas** as bright, round masses near the skull or brain base; **pituitary tumours** mainly in sagittal and axial views near the centre of the skull.
- **No-tumour scans look different in style** (contrast and brightness), which is a warning sign: a model could learn the scan style instead of the tumour.



Figure 3. Average brightness per class (sampled images). Glioma scans are the darkest on average and no-tumour scans the brightest.

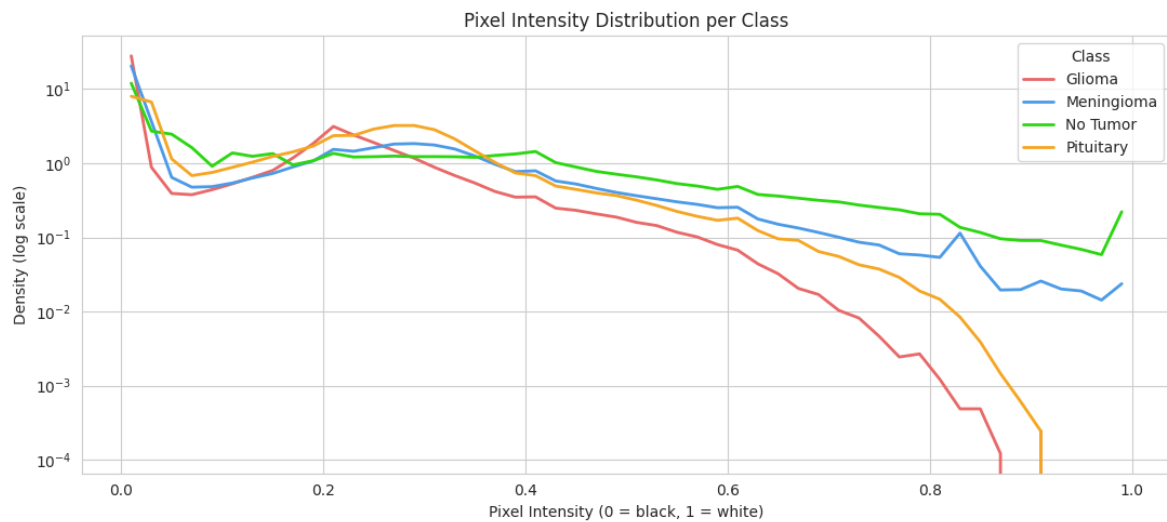


Figure 4. Pixel intensity distribution per class (log scale). MRI backgrounds are black, producing a large peak at low intensities.

The classes differ in their intensity profile, which supports normalising pixel values and adding brightness and contrast augmentation. It also explains why confusion matrices and misclassified images were studied in addition to accuracy.

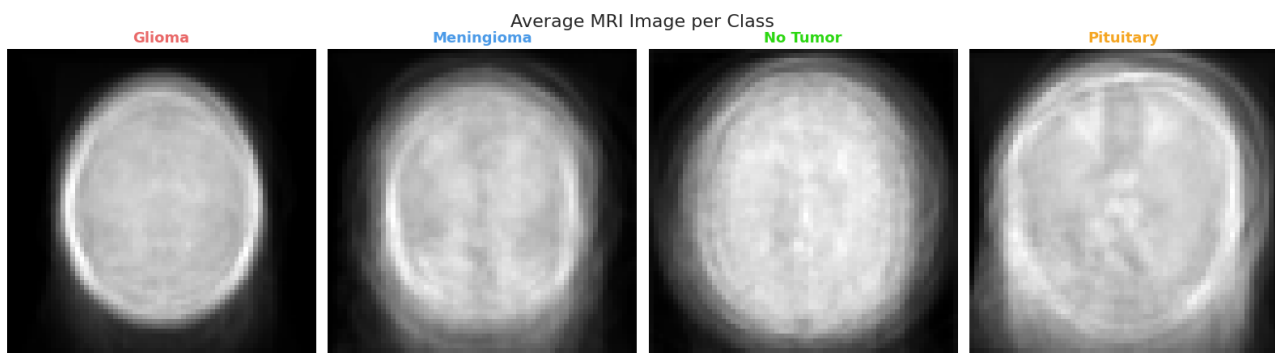


Figure 5. Pixel-wise average image of each class. Averages look like blurred brains, so class differences are not visible globally.

5. Methodology

Preprocessing, augmentation, model architectures and training strategy.

Pipeline overview

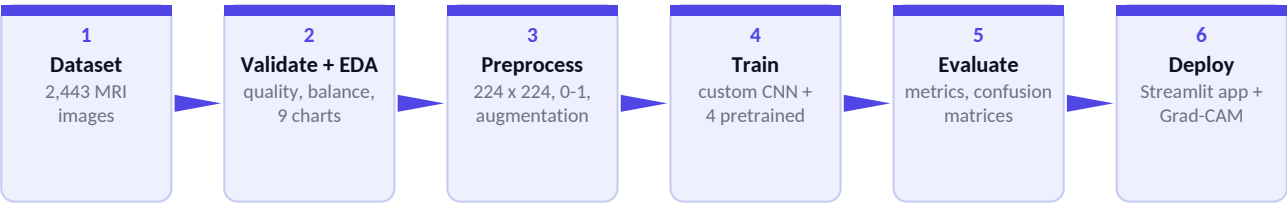


Figure. Overview of the project pipeline.

5.1 Preprocessing

Table 6. Preprocessing pipeline.

Step	Action	Reason
1	Decode every image to 3-channel RGB	Pretrained networks expect 3 channels
2	Resize to 224 x 224 (bilinear)	Standard input size of the pretrained models
3	Scale pixel values to the 0-1 range	Stable, faster training
4	tf.data pipeline: parallel loading, in-memory cache (uint8), batching (32), prefetch	Reads each file from disk only once; fast epochs
5	Shuffle training data only	Validation and test order stays aligned with labels

5.2 Data augmentation

Augmentation creates slightly changed copies of the **training** images on the fly, which shows the model more variety and reduces overfitting. Validation and test images are never augmented, so evaluation stays fair.

Table 7. Augmentation settings (training data only).

Transformation	Setting
Random flip	Horizontal and vertical
Random rotation	Up to 5% of a full turn (about 18 degrees)
Random zoom	Up to 15%
Random translation (shift)	Up to 10% in both directions
Random brightness	Up to 20%
Random contrast	Up to 20%

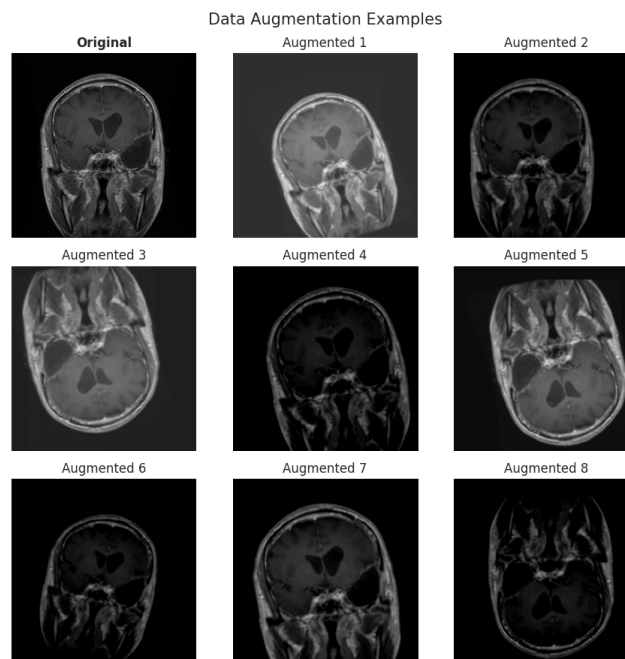


Figure 6. One validation image and eight random augmented versions.

5.3 Handling class imbalance

Balanced class weights are passed to the training loop so that smaller classes receive a larger influence on the loss.

Table 8. Class weights computed from the training set.

Class	Glioma	Meningioma	No Tumor	Pituitary
Class weight	0.751	1.184	1.265	0.967

5.4 Custom CNN (built from scratch)

The custom network stacks four convolutional blocks of increasing width. Each convolution is followed by Batch Normalization (stabilises learning) and ReLU. Every block ends with Max Pooling (halves the spatial size) and Dropout (randomly switches off neurons to reduce overfitting). Global Average Pooling replaces a large flatten layer, which keeps the model small: **1,232,612 parameters** in total.

Table 9. Custom CNN architecture.

Stage	Layers	Output shape
Input	RGB image, pixels 0-1	224 x 224 x 3
Block 1	Conv 3x3 (32) + BN + ReLU, MaxPool 2x2, Dropout 0.25	112 x 112 x 32
Block 2	2 x [Conv 3x3 (64) + BN + ReLU], MaxPool 2x2, Dropout 0.25	56 x 56 x 64
Block 3	2 x [Conv 3x3 (128) + BN + ReLU], MaxPool 2x2, Dropout 0.25	28 x 28 x 128
Block 4	2 x [Conv 3x3 (256) + BN + ReLU], MaxPool 2x2, Dropout 0.25	14 x 14 x 256
Pooling	Global Average Pooling	256
Classifier	Dense 256 (ReLU), Dropout 0.5	256
Output	Dense 4 (Softmax)	4 class probabilities

5.5 Transfer learning

Four networks pretrained on ImageNet were reused, each with a new classification head. A small *Rescaling* layer inside each model converts the 0-1 pixels into the range the pretrained network was trained with (-1 to 1 for MobileNetV2, ResNet50V2 and InceptionV3; 0 to 255 for EfficientNetB0). The new head consists of Global Average Pooling, Batch Normalization, Dropout (0.4), Dense 256 (ReLU), Dropout (0.3) and a 4-way Softmax.

Table 10. Two-phase transfer learning strategy.

Phase	What is trained	Learning rate	Max epochs
1. Feature extraction	New head only; pretrained base frozen	1e-3	10
2. Fine-tuning	Top 30 layers of the base plus the head; Batch Normalization layers stay frozen	1e-5	10

Table 11. Pretrained architectures used.

Model	Role and characteristics	Parameters
MobileNetV2	Very light network built for mobile devices; fast inference	2.59 M
ResNet50V2	50 layers with residual (shortcut) connections; powerful but heavy	24.10 M
InceptionV3	Filters of several sizes in parallel capture small and large patterns	22.34 M
EfficientNetB0	Compound scaling of depth, width and resolution; good size-to-accuracy ratio	4.38 M

ResNet50V2 was used instead of the original ResNet50 because its input scaling can be built from standard layers, which keeps the saved .h5 file simple to reload.

5.6 Training configuration

Table 12. Training configuration shared by all models.

Setting	Value
Optimizer / loss	Adam / categorical cross-entropy
Batch size	32
Epochs (maximum)	Custom CNN: 30. Transfer learning: 10 (head) + 10 (fine-tuning)
EarlyStopping	Monitor validation loss, patience 5, restore best weights
ModelCheckpoint	Save only the best model (lowest validation loss) as .h5
ReduceLROnPlateau	Factor 0.3, patience 2, minimum learning rate 1e-7
Class weights	Balanced (see Section 5.3)
Random seed	42 for Python, NumPy and TensorFlow
Hardware	Google Colab GPU runtime

5.7 Evaluation protocol

- **Model selection on validation data.** The best model is chosen by validation macro F1-score (ties broken by smaller file size). The test set is used once, for the final report.
- **Metrics.** Accuracy, macro precision, macro recall, macro F1-score, per-class F1, confusion matrix and training curves.
- **Tumour sensitivity.** Of all scans that truly contain a tumour (glioma, meningioma or pituitary), the share that the model did *not* label as "No Tumor". Missing a tumour is the most serious mistake in a medical setting, so this metric is reported next to accuracy.
- **Efficiency.** Parameter count, file size and inference time per image.

6. Results and Model Comparison

All five models, evaluated with the same code on the same data.

6.1 Performance comparison

Table 13. Validation and test metrics (precision, recall and F1 are macro-averaged). Rows sorted by validation F1.

Model	Val acc.	Val F1	Test acc.	Test prec.	Test recall	Test F1	Tumour sens.
ResNet50V2	0.8984	0.8959	0.8618	0.8696	0.8576	0.8546	0.9898
InceptionV3	0.8825	0.8775	0.8415	0.8490	0.8418	0.8358	0.9848
MobileNetV2	0.8466	0.8404	0.8252	0.8391	0.8259	0.8232	0.9949
Custom CNN	0.7948	0.7618	0.7846	0.8101	0.7844	0.7530	0.9188
EfficientNetB0	0.6773	0.6608	0.6992	0.7972	0.6786	0.6874	0.9949



Figure 7. Test-set accuracy, precision, recall and F1-score of all models.

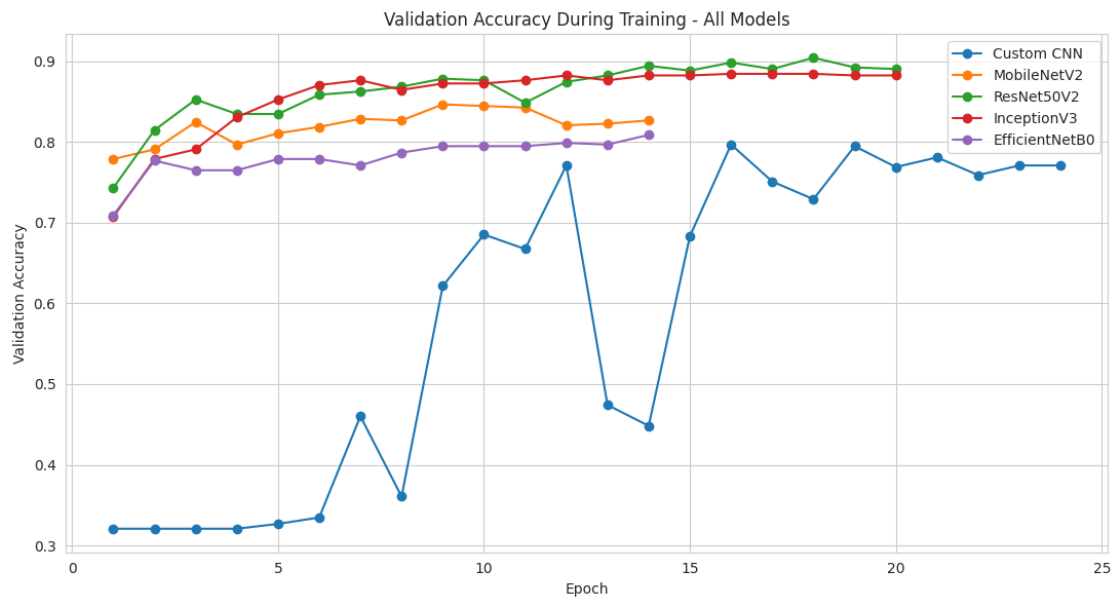


Figure 8. Validation accuracy during training. The pretrained models converge within a few epochs; the custom CNN starts slowly and fluctuates.

The custom CNN's validation accuracy stayed at about 0.32 for its first five epochs, which matches the glioma share of the validation set (161 of 502 images = 0.321). In other words, the network initially predicted a single class and needed several epochs before it began to learn useful features.

6.2 Per-class behaviour

Table 14. Per-class F1-score on the test set.

Model	Glioma	Meningioma	No Tumor	Pituitary
ResNet50V2	0.950	0.761	0.879	0.828
InceptionV3	0.917	0.726	0.870	0.831
MobileNetV2	0.889	0.710	0.851	0.844
Custom CNN	0.848	0.452	0.807	0.904
EfficientNetB0	0.731	0.483	0.824	0.713

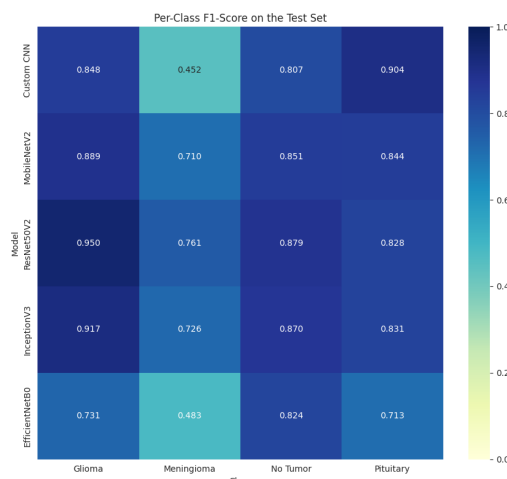


Figure 9. Per-class F1-score heatmap. Meningioma is the weakest class for every model.

6.3 Selected model: ResNet50V2

Table 15. Classification report of ResNet50V2 on the test set.

Class	Precision	Recall	F1-score	Support
Glioma	0.95	0.95	0.95	80
Meningioma	0.86	0.68	0.76	63
No Tumor	0.95	0.82	0.88	49
Pituitary	0.72	0.98	0.83	54
Accuracy			0.86	246

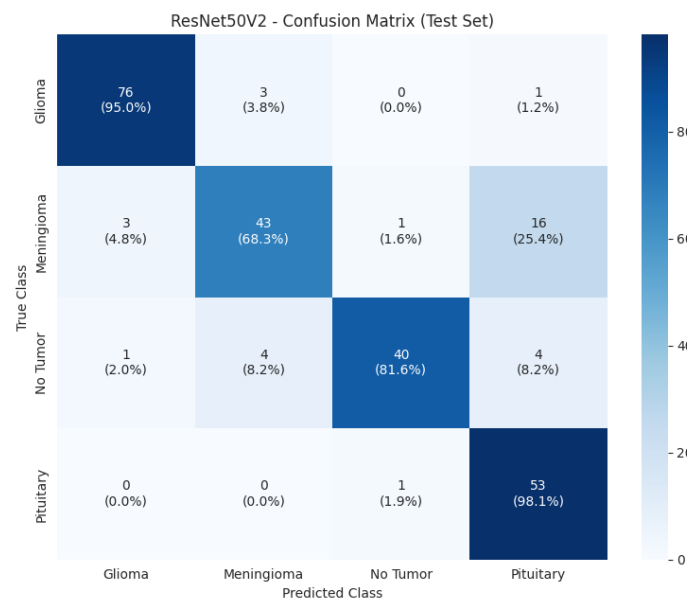


Figure 10. Confusion matrix of ResNet50V2 on the test set (counts and row percentages).

- **Glioma and pituitary are recognised well** (recall 95% and 98%).
- **Meningioma is the main weakness:** 16 of 63 scans (25.4%) were predicted as pituitary. The notebook did not investigate the cause; similar appearance in some viewing planes is a plausible factor.
- **Only 2 of 197 tumour scans were labelled "No Tumor"** (1 meningioma and 1 pituitary), giving a tumour sensitivity of 98.98%.
- **Pituitary precision is lowest (0.72)** because it absorbs the meningioma errors.

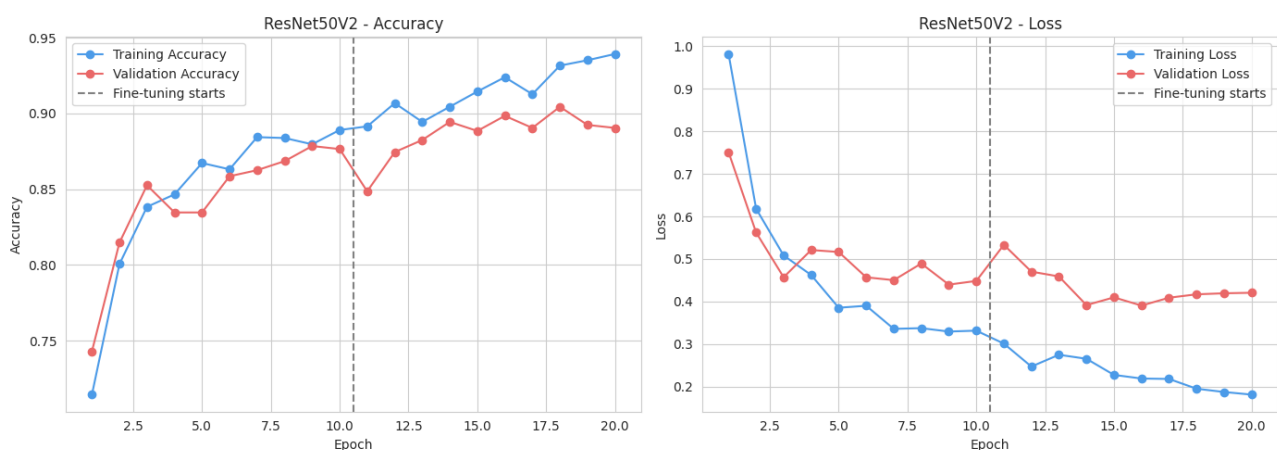


Figure 11. Training and validation curves of ResNet50V2. The dashed line marks the start of fine-tuning.

Training accuracy ends near 0.94 and validation accuracy near 0.89, a gap of about five points: a mild degree of overfitting that is controlled by dropout, augmentation and early stopping. Validation loss plateaus around 0.41, and fine-tuning gave a visible but modest improvement over the frozen-base phase.

6.4 Comparison of all confusion matrices

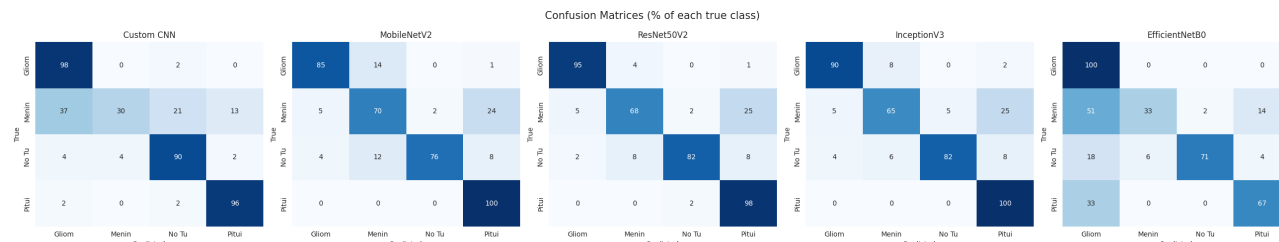


Figure 12. Row-normalised confusion matrices of all models (percent of each true class).

- All three strong pretrained models (MobileNetV2, ResNet50V2, InceptionV3) misclassify roughly **24-25% of meningioma scans as pituitary**, so the confusion reflects a genuinely hard distinction in this data rather than a single model's flaw.
- The **custom CNN** recognises only 30% of meningiomas and sends 37% of them to glioma.
- EfficientNetB0** predicts glioma far too often (for example 51% of meningiomas and 33% of pituitary tumours), which explains its low accuracy. Its tumour sensitivity stays high only because it rarely predicts "No Tumor".

6.5 Efficiency

Table 16. Model size and speed. 'Weights only' is the approximate size when stored in 32-bit floats without optimizer state.

Model	Parameters (M)	Weights only (MB)	Checkpoint (MB)	Inference (ms/img)	Training (min)
ResNet50V2	24.10	92	206.4	3.04	10.2
InceptionV3	22.34	85	129.2	2.85	10.6
MobileNetV2	2.59	10	24.3	1.21	6.4
Custom CNN	1.23	5	14.2	2.16	11.1
EfficientNetB0	4.38	17	31.0	2.60	7.9

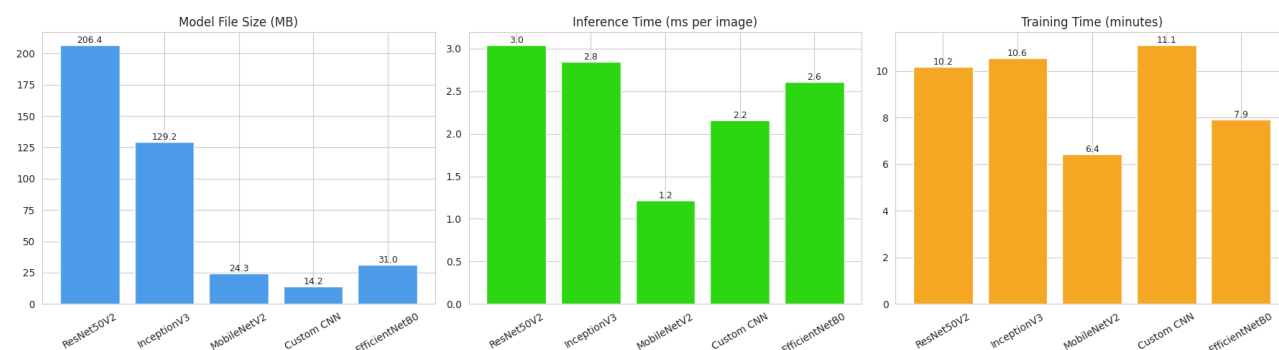


Figure 13. Model file size, inference time and training time.

Inference time is measured per image over one pass of the test set on the Colab GPU. The checkpoint saved during training also contains the optimizer state, which is why it is larger than the weights alone (see Section 7.4).

6.6 Misclassified examples

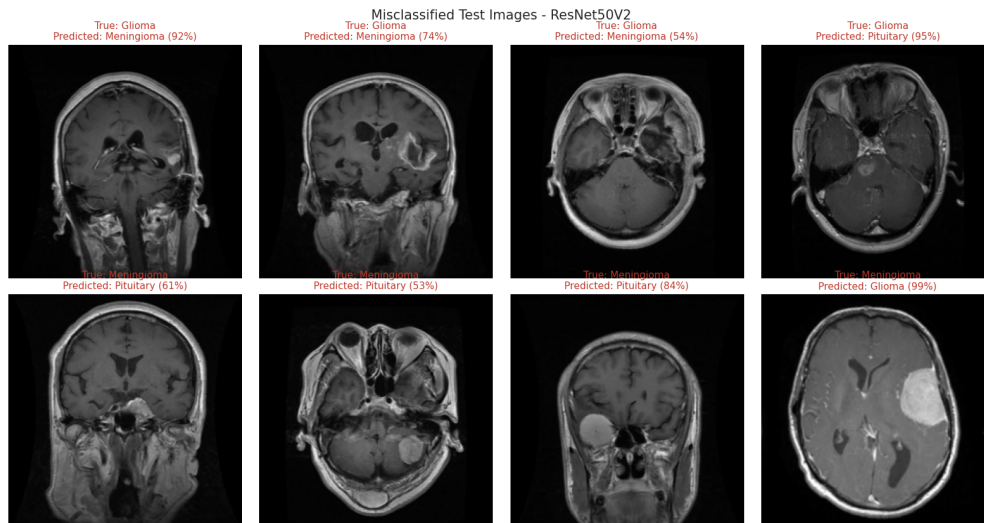


Figure 14. Misclassified test images of ResNet50V2, with true label, prediction and confidence.

Some errors are made with high confidence (for example, a meningioma predicted as glioma at 99%). Confidence therefore should not be read as a guarantee of correctness, which is why the application also reports the runner-up class and flags close calls.

6.7 Model selection decision

Decision

ResNet50V2 was selected: it has the highest validation F1-score (0.8959) and also the best test accuracy (86.18%), precision, recall and F1-score. **MobileNetV2** is the recommended alternative where size and speed matter: it is about 8 times smaller in weights, the fastest at inference, and has the highest tumour sensitivity (99.49%), at a cost of about 3.7 points of accuracy.

7. Application and Deployment

The Streamlit web application and how it is deployed.

7.1 Inference flow

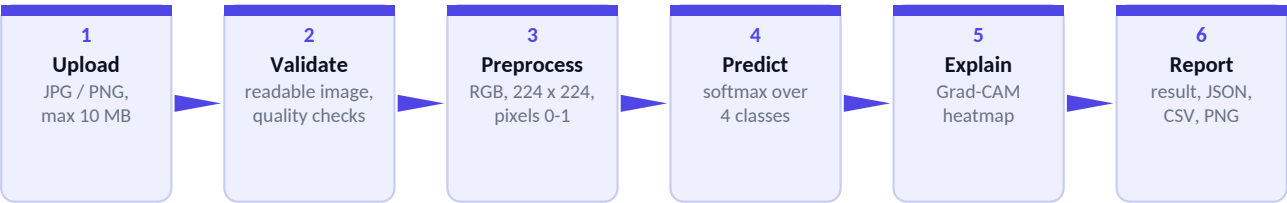


Figure. Steps the application performs for every uploaded scan.

Preprocessing in the application reproduces the training pipeline exactly (same bilinear resize and rounding), which avoids small train-serve differences.

7.2 Features

Table 17. Application features.

Area	Capability
Single-scan analysis	Predicted class, confidence with High / Medium / Low level, runner-up class, top-2 margin, inference time and probabilities of all four classes.
Explainability	Grad-CAM heatmap with Original, Heatmap and Overlay views and an adjustable strength. It highlights influential regions; it is not a tumour outline.
Batch mode	Up to 25 images at once with a summary, class distribution chart, results table and CSV download.
Model Insights	Model card (architecture, parameters, size), the comparison table and charts, metric explanations and known limitations.
Quality checks	Resolution, grayscale and file-size checks. Colour images, very small images and low-confidence or close-call results trigger warnings.
Exports	Report as JSON, heatmap as PNG, batch results as CSV; session history of the last 20 analyses.
Interface	Premium dark and light themes, tabs, responsive layout and a clear medical disclaimer.
Robustness	Friendly messages for oversize files, invalid images, missing or mismatched model files and prediction errors; model and results are cached for speed.

7.3 Repository structure

```
Brain-Tumor-MRI-Classification/
|-- .streamlit/config.toml      # theme colour, 10 MB upload limit
|-- images/                   # charts saved by the notebook
|-- models/
|   |-- best_model.h5          # model used by the app
|   |-- class_names.json      # class order of the outputs
|-- .gitignore
|-- README.md
|-- app.py                    # Streamlit application
|-- requirements.txt
`-- Brain_Tumor_MRI_Classification.ipynb # full project notebook
```

The dataset (data/) and the other trained models are excluded through .gitignore because of GitHub's file-size limit; only the two files the application needs are committed.

7.4 Deployment

The application is deployed on Streamlit Community Cloud from the GitHub repository (main file app.py). Two practical constraints shaped the deployment:

Table 18. Deployment issues and resolutions.

Issue	Cause	Resolution
Model file larger than GitHub's 100 MB limit	The ResNet50V2 checkpoint (206 MB) stores optimizer state in addition to the weights (about 92 MB).	Re-save the model without optimizer state (load with compile=False, then save). Verified: file size drops to about 92 MB and predictions are identical (maximum difference 0.0).
Dependency installation failed on the cloud	The cloud environment used Python 3.14, for which no stable TensorFlow release has wheels yet (latest stable: 2.21.0 supports Python 3.10 to 3.13).	Select Python 3.12 in the app's advanced settings (the Python version cannot be changed after deployment, so the app is redeployed) and pin the TensorFlow version.
Memory on the free tier	TensorFlow plus a 92 MB model is close to the roughly 1 GB available.	Use MobileNetV2 as the deployed model if the app runs out of memory.

Recommended requirements.txt

```
tensorflow-cpu==2.21.0 # use the same version as in the training environment
streamlit>=1.50
numpy
pandas
matplotlib
Pillow
```

Run locally

```
python -m venv .venv
.venv\Scripts\activate # macOS/Linux: source .venv/bin/activate
pip install -r requirements.txt
streamlit run app.py
```

8. Limitations, Risks and Responsible Use

Read before interpreting or reusing the results.

Table 19. Known limitations.

Limitation	Impact	Mitigation / next step
About 19% (validation) and 23% (test) of images share a scan ID with training images	Scores are probably optimistic for new patients	Patient-level split; evaluate on an independent dataset
Small test set (246 images)	One error changes accuracy by about 0.4 points; small differences between models are not meaningful	Larger test set; cross-validation; confidence intervals
Scan-style differences between classes	A model may partly learn scan appearance instead of the tumour	Intensity normalisation; multi-centre data; Grad-CAM review
Meningioma recall of 68%	About a quarter of meningiomas are mistaken for pituitary tumours	More meningioma data; targeted augmentation; ensembling
No clinical validation	Behaviour on other hospitals, scanners and protocols is unknown	External validation with radiologist review
Overconfident errors	Confidence is not a guarantee of correctness	Probability calibration; show runner-up and close-call warnings

Responsible use

This system is a student project built to demonstrate deep learning on medical images. It is **not** a medical device, has not been clinically validated and must never be used for diagnosis or treatment decisions. Every prediction must be reviewed by a qualified radiologist or doctor. Uploaded images are processed in memory and are not stored by the application.

Future work

- Patient-level data splitting and external validation to obtain a trustworthy estimate of real-world performance.
- Ensembling the best models, and targeted work to improve meningioma recognition.
- Probability calibration so that confidence values better reflect true accuracy.
- Extending Grad-CAM to a more detailed explanation view and adding a radiologist feedback loop.
- Packaging the model as an API (for example with a container) for integration into hospital workflows.

9. How to Reproduce the Project

From a clean machine to a running application.

Table 20. Reproduction steps.

Step	Action
1	Clone the repository and create a virtual environment (Python 3.12 recommended).
2	Place the dataset in a folder named <code>data</code> next to the notebook, containing <code>train</code> , <code>valid</code> and <code>test</code> (each with <code>glioma</code> , <code>meningioma</code> , <code>no_tumor</code> and <code>pituitary</code> folders).
3	Open the notebook on a GPU runtime (for example Google Colab). If running on Colab, upload the zipped dataset to Drive, mount Drive and unzip it to <code>/content/data</code> .
4	Set <code>QUICK_RUN = True</code> for a fast end-to-end test, then set it back to <code>False</code> and run all cells. Internet access is needed once to download the ImageNet weights.
5	Training saves each best model to <code>models/</code> , writes charts to <code>images/</code> and copies the winner to <code>models/best_model.h5</code> together with <code>class_names.json</code> .
6	Re-save <code>best_model.h5</code> without optimizer state if it exceeds 100 MB, then run <code>streamlit run app.py</code> .
7	Push to GitHub and deploy on Streamlit Community Cloud with Python 3.12.

Notebook structure

Table 21. Structure of the project notebook.

Section	Content
1 - 3	Know your data, variables and data wrangling (quality, imbalance, split overlap)
4	Nine EDA charts (univariate, bivariate, multivariate) with insights
5	Preprocessing, augmentation, <code>tf.data</code> pipeline and class weights
6	Custom CNN and four transfer learning models: training and evaluation
7 - 8	Evaluation overview, model comparison and best-model selection
9 - 11	Saving the model, single-image inference and Streamlit application

Appendix A. Glossary

Terms used in this document.

Term	Meaning
CNN	Convolutional Neural Network: a network that learns visual patterns with small sliding filters.
Transfer learning	Reusing a network pretrained on a large dataset (ImageNet) and adapting it to a new task.
Fine-tuning	Unfreezing the top layers of a pretrained network and training them with a very small learning rate.
Batch Normalization	Normalises layer outputs during training to make learning faster and more stable.
Dropout	Randomly switches off neurons during training to reduce overfitting.
Softmax	Converts the network's outputs into four probabilities that add up to 1.
Macro average	Metric computed per class and averaged, so every class counts equally regardless of its size.
F1-score	Harmonic mean of precision and recall.
Tumour sensitivity	Share of scans with a real tumour that were not labelled 'No Tumor'.
Confusion matrix	Table of true classes against predicted classes; the diagonal shows correct predictions.
Grad-CAM	Technique that uses gradients of the last feature maps to show which image regions influenced a prediction.
Class weights	Larger loss weights for under-represented classes to counter imbalance.
EarlyStopping	Stops training when validation loss stops improving.

Appendix B. Technology stack

Layer	Tools
Deep learning	TensorFlow / Keras (custom CNN, MobileNetV2, ResNet50V2, InceptionV3, EfficientNetB0)
Data and metrics	NumPy, pandas, scikit-learn (metrics, class weights), Pillow
Visualisation	Matplotlib, Seaborn (notebook); Altair (application charts)
Application	Streamlit (Community Cloud), Grad-CAM implemented with TensorFlow GradientTape
Training environment	Google Colab GPU runtime
Version control	Git and GitHub

Repository: <https://github.com/13msrajput/Brain-Tumor-MRI-Classification>

Live application: <https://brain-tumor-mri-classification-system.streamlit.app>